THE CHINESE ECONOMY

A JOURNEY INTO HISTORY, ECONOMICS, AND POLICY

Lecture 6: Inside the sectors

TODAY

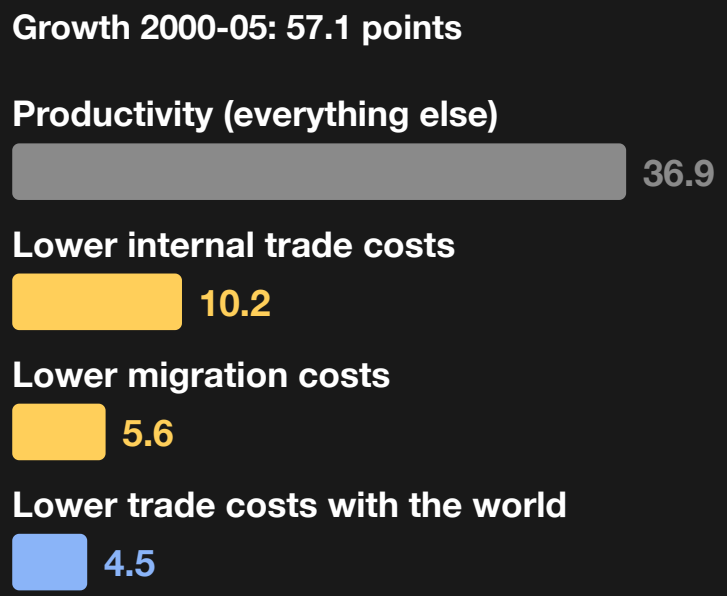
- **Finishing lecture 5**: Khanna et al., and "gone or blocked?"
- **Inside the sectors**: what is in the grey bar, and the elephant in it
- **A model with many firms**: two inputs, two wedges, two TFPs
- **The SOE Lab**: three questions, on the screen
- **One paper, in seven steps**: Hsieh and Klenow (2009)

FIRST: THE END OF LECTURE 5

- Khanna, Liang, Mobarak and Song (2025): the whole paper, quickly, from the title
- Gone or blocked? City wages rise fast: are the extra farm workers gone, or kept on the farm?
- That closes the world **between** sectors: the farm, the city, and the wall between them

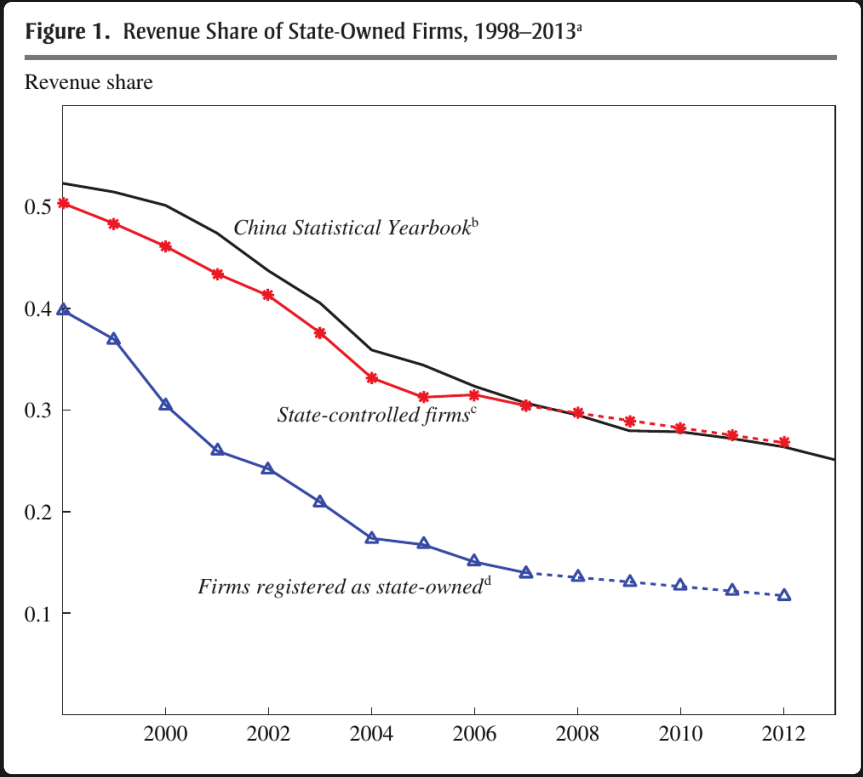
WHERE DID THE GROWTH COME FROM?

TOMBE AND ZHU (2019), LAST WEEK



- Two thirds of the growth sits in the **grey bar**: "productivity, everything else"
- What is inside? The authors: changes "in the average efficiency or average capital intensity of the firms [...] or the changes in **capital allocation among these firms**"
- So far we have been **silent** on what happens **inside** a sector. Today we open it

INSIDE THE SECTORS, AN ELEPHANT: THE STATE



Hsieh and Song (2015), Fig. 1: share of industrial revenue of state firms, 1998-2013. Red: firms controlled by the state, however registered

- 1978: state firms made **77%** of industrial output. 1996: **33%**
- 1998: still about **half** of industrial revenue, and **40.6%** of them were losing money
- From about 1993, more than **30 million** workers of state firms were laid off
- By 2005, the state's share was down to about **30%**

QUESTION

EVERY ECONOMY LEAVING THE PLAN

What do you do with the state firms?

- **Close** the ones that lose money, and let their workers and capital go elsewhere
- **Sell** them: privatise, as Russia and Central Europe did in the 1990s
- **Keep** them, but make them behave like firms that must earn a profit
- China's slogan: **"Grasp the large, let go of the small"**. Close or sell the small ones; merge the large ones into groups the state controls
- Tomorrow: what China did, and what it was worth. **Today**: how much does it cost to keep capital and workers in the wrong firms?

FROM TWO SECTORS TO MANY FIRMS

	Lectures 4-5: the hukou	**Today: the state firms**
Units	two sectors, farm and city	many **firms** in one industry
Inputs	labour	**capital and labour**
Wedges	one: τ, the cost of moving	two per firm: τ_K (credit), τ_L (pressure to keep workers)
Why more is worth less	$\gamma < 1$: each extra worker adds less	twice the inputs, twice the output; but to sell it a firm must **lower its price**: revenue only **×1.6**
Efficiency	$MPL_F = MPL_C$	the **same** MRPL and MRPK in every firm
What we look at	one gap	a **distribution** across firms

THE STORY OF THE MODEL

Firm i produces with capital and labour: $Y_i = A_i\, K_i^{\alpha} L_i^{1-\alpha},\ 0 < \alpha < 1$

Each firm sells its own product; to sell more it must lower its price P_i

It pays $(1 + \tau_{L,i})\, w$ per worker and $(1 + \tau_{K,i})\, r$ per unit of capital. Cheap credit: $\tau_K < 0$. Pressure to keep workers: $\tau_L < 0$

It hires until the last worker and the last unit of capital pay for themselves:

$$MRPL_i \propto \frac{P_i Y_i}{L_i} = (1 + \tau_{L,i})\, w \qquad MRPK_i \propto \frac{P_i Y_i}{K_i} = (1 + \tau_{K,i})\, r$$

So we can **see** them in the data: **revenue per worker** and **revenue per unit of capital**, firm by firm. Without wedges they are the same everywhere

TWO TFPS

TFPQ: output in **physical units** per unit of input $TFPQ_i = A_i = \frac{Y_i}{K_i^{\alpha} L_i^{1-\alpha}}$

TFPR: output in **yuan** per unit of input $TFPR_i = P_i A_i = \frac{P_i Y_i}{K_i^{\alpha} L_i^{1-\alpha}}$

> Two steel mills. Mill B makes **twice** the steel of mill A with the same inputs. Without wedges, B gets more capital and workers and sells more steel; its price falls. It stops growing when one more yuan of inputs earns **the same** as at mill A

So without wedges **TFPR is the same** in every firm. With wedges:

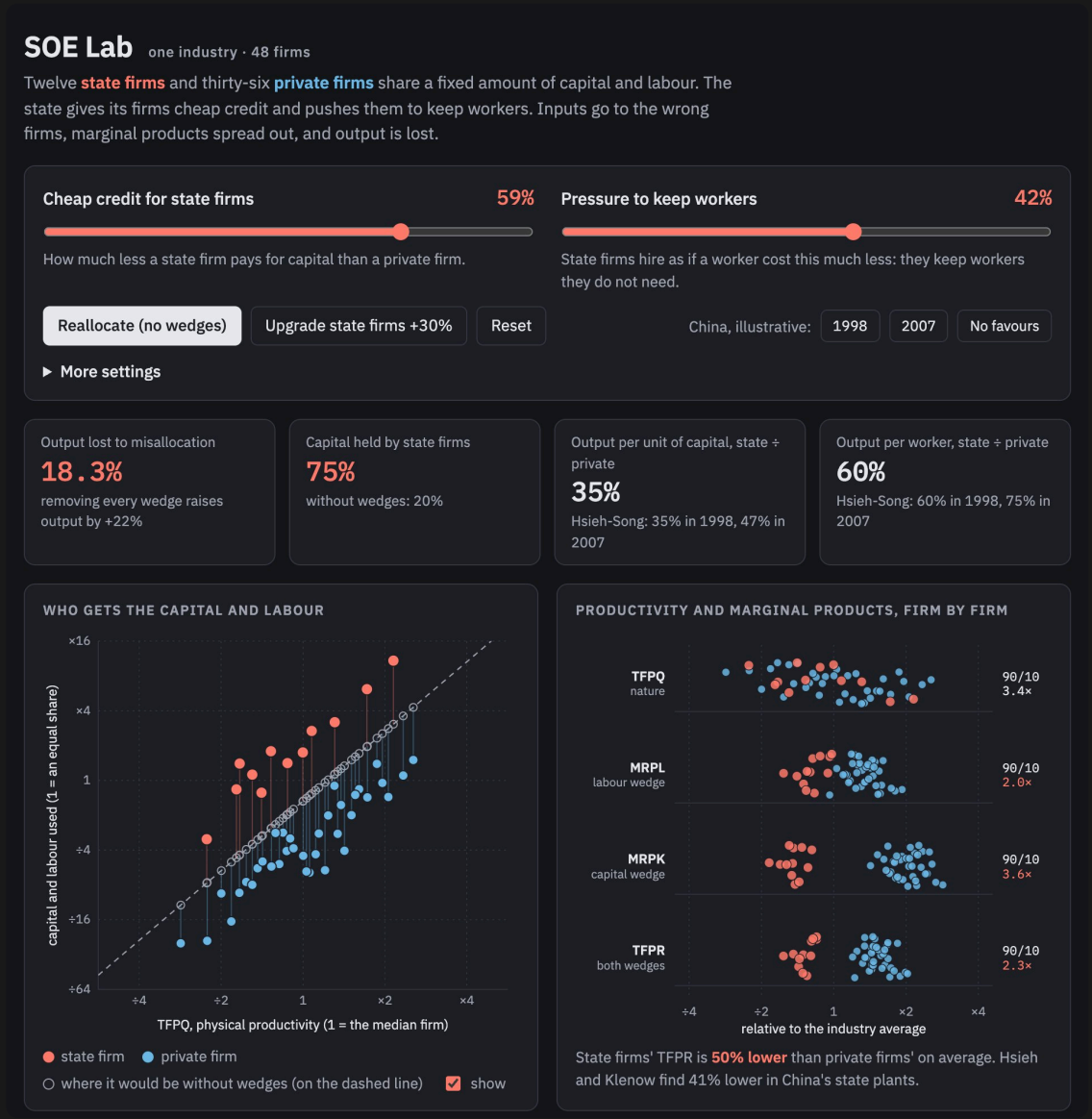
$$TFPR_i \propto (1 + \tau_{K,i})^{\alpha} (1 + \tau_{L,i})^{1-\alpha}$$

Differences in **TFPQ** are technology and management: a wedge changes how much a firm uses and sells, not what it makes with each unit. Differences in **TFPR** are the **wedges**

RULES OF THE WORLD

- Every firm faces the **same** wage and interest rate; only the wedges differ
- All firms of an industry use the **same** mix: the same α, taken from the same industry in the United States
- All firms charge the **same** markup. Demand, with $\sigma = 3$:
  $Y_i = Y\left(P_i/P\right)^{-\sigma}$, so price $= 1.5\times$ marginal cost
  P, Y: price and output of the industry; $1.5 = \sigma/(\sigma - 1)$
- So **everything** that makes revenue per input differ across firms is counted as a **wedge**, including errors in the data

TRY IT: THE SOE LAB



Open the lab · 12 state firms, 36 private firms, the same capital and labour

SOE LAB 1

SOE LAB

Press "Reallocate". Which rows of the right-hand chart move, and which do not? Why?

- **TFPQ does not move**: it is what each firm can do, its technology and management
- MRPL, MRPK and TFPR **collapse onto one line**: they differed only because of the wedges
- State firms **shrink**, the good private firms **grow**: on the left, every firm moves to the dashed line
- Same capital, same workers: output is about **22%** higher

SOE LAB 2

SOE LAB

Click "1998", then "2007". What changed in the state firms, and what did not?

- **Output per worker**: from 60% to 75% of private firms. Less pressure to keep workers: the state firms that survived shed about 40% of their workers
- **Output per unit of capital**: from 35% to only 47%. Credit stayed cheap, even for the state firms that were privatised
- In the lab, output lost falls from about **18%** to about **9%**. Tomorrow: what the real reform was worth

SOE LAB 3

SOE LAB

Press "Upgrade state firms +30%". Why is the net gain smaller than the direct gain?

- Direct gain: the state firms produce more with what they have, **+15%**
- But they now pull in more capital and workers, away from private firms where they produce **more**: **−8%**
- Net: **+7%**. Reset, press "Reallocate", then upgrade again: the drag disappears
- Making a **favoured** firm better is not a free lunch. Tomorrow: in China's data the drag can exceed the gain

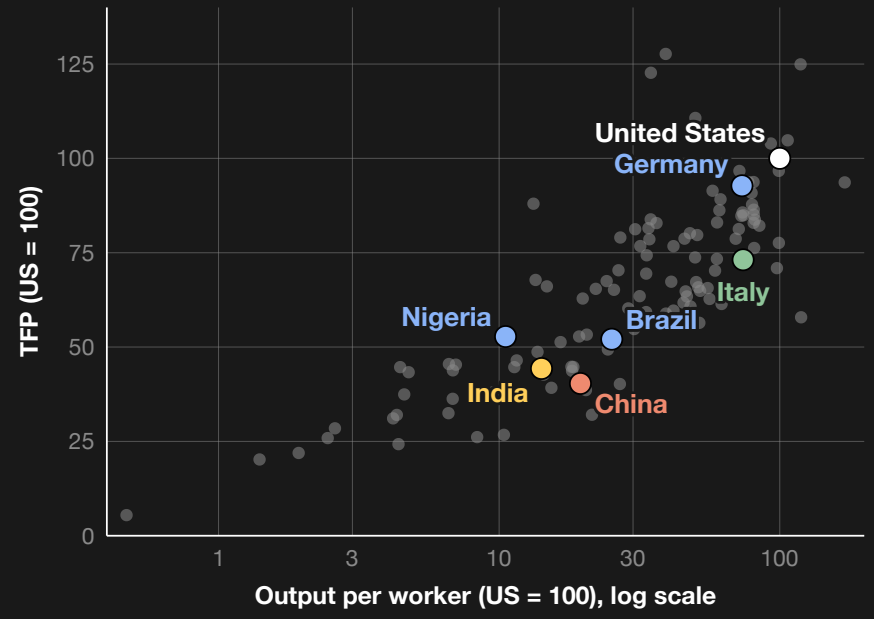
MISALLOCATION AND MANUFACTURING TFP IN CHINA AND INDIA

Chang-Tai Hsieh, Peter J. Klenow

Quarterly Journal of Economics, 2009

- Read the title as last week: **misallocation** is the cause, **TFP** is the effect, **manufacturing** in two big poor countries is the place

MOTIVATION



118 countries in 2019. TFP: output with the same capital and the same workers (and their schooling). Penn World Table 10.01

- Poor countries produce much less **with the same capital and workers**: their **TFP** is lower. China: output per worker **19%** of the US, TFP **40%**
- Inside a narrow industry, firms differ a lot. If capital and workers do not go to the best ones, the whole industry is less productive
- In China and India the **state** decides much: who gets credit, who may grow, who must keep workers
- Is part of the gap with the rich countries **not technology, but allocation**?

QUESTION

HSIEH AND KLENOW (2009) · 2. QUESTION

How much higher would manufacturing TFP be in China and India if capital and labour were allocated across firms as well as in the United States?

DATA

- **China**: the Annual Survey of Industrial Production, 1998-2005. All state firms, and all other firms with more than 5 million yuan of revenue (about $600,000): from over 100,000 firms in 1998 to over 200,000 in 2005
- **India**: the Annual Survey of Industries, 1987-1994. **United States**: the Census of Manufactures, 1977-1997
- For each firm: its industry (4 digits), value added, wage bill, capital, ownership

QUESTION

HSIEH AND KLENOW (2009) · 3. DATA

What if the data are wrong?

- An error in a firm's revenue or capital looks **exactly like a wedge**: the model counts it as misallocation
- So the measured gains are **too large** if the data are noisy, and noisier data in China and India would explain part of the gap with the US
- **SOE Lab**: open "Errors in the data". The same error for all firms, then a bigger one for state firms. What happens to the loss we **measure**? And to the true one?
- Their check: pure noise would not depend on **who owns the firm**. It does: state plants have about **40% lower** TFPR

EMPIRICAL STRATEGY

HSIEH AND KLENOW (2009) · 4. STRATEGY

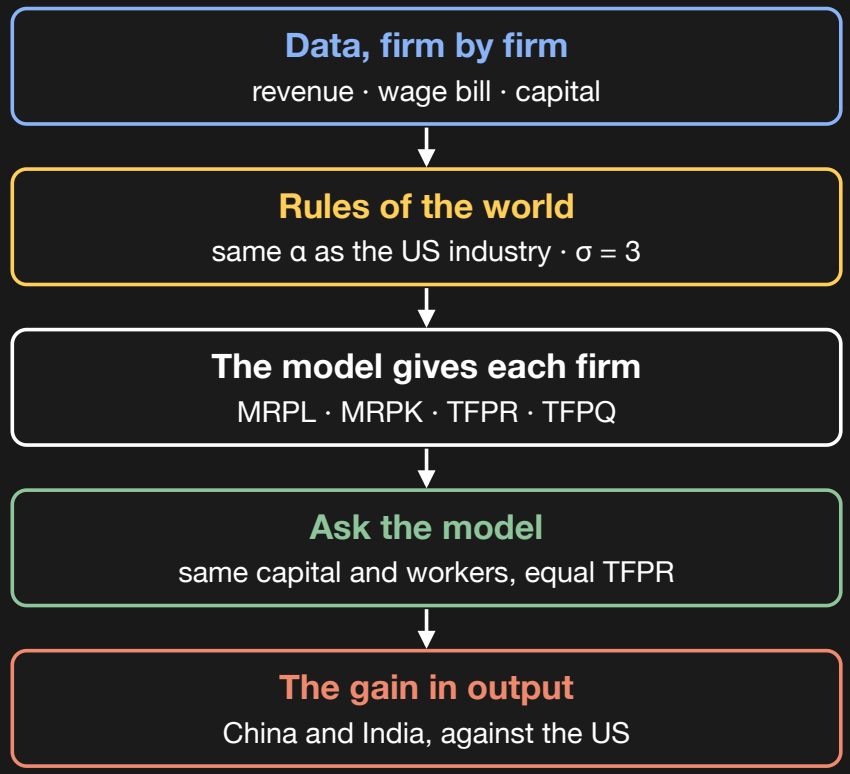
No lottery of nature this time: the key numbers come out of the **model**, simulated with **real data**

Firm by firm: revenue per worker and per unit of capital give **MRPL** and **MRPK**; together, **TFPR**

Prices are not in the data. Revenue and demand ($\sigma = 3$) give quantity and price, so **TFPQ**: what each firm can make

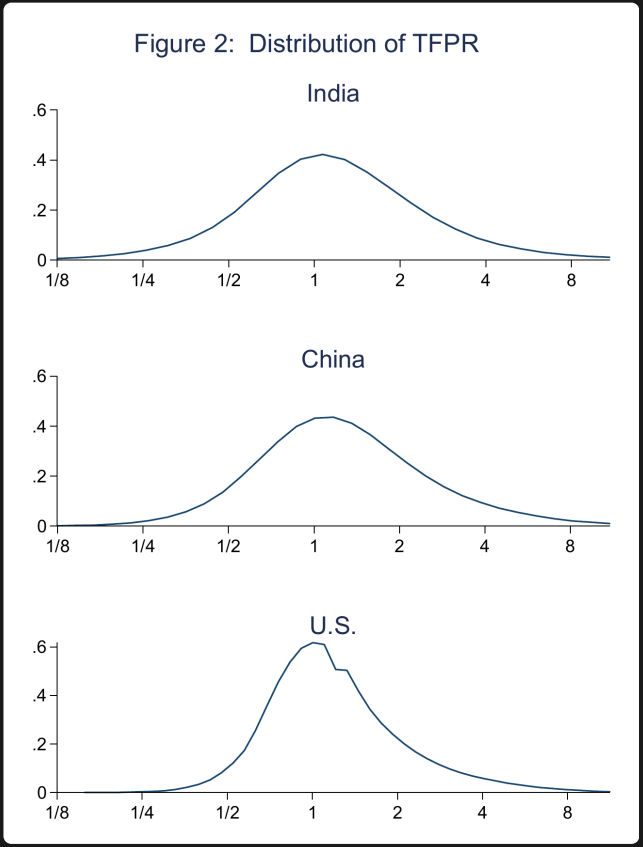
Ask the model: the **same** capital and workers, but **equal TFPR** in every firm. How much more output?

Do it for China, India and the US, and compare: the gain of reaching **US efficiency**



RESULTS: HOW SPREAD OUT

HSIEH AND KLENOW (2009) · 5. RESULTS

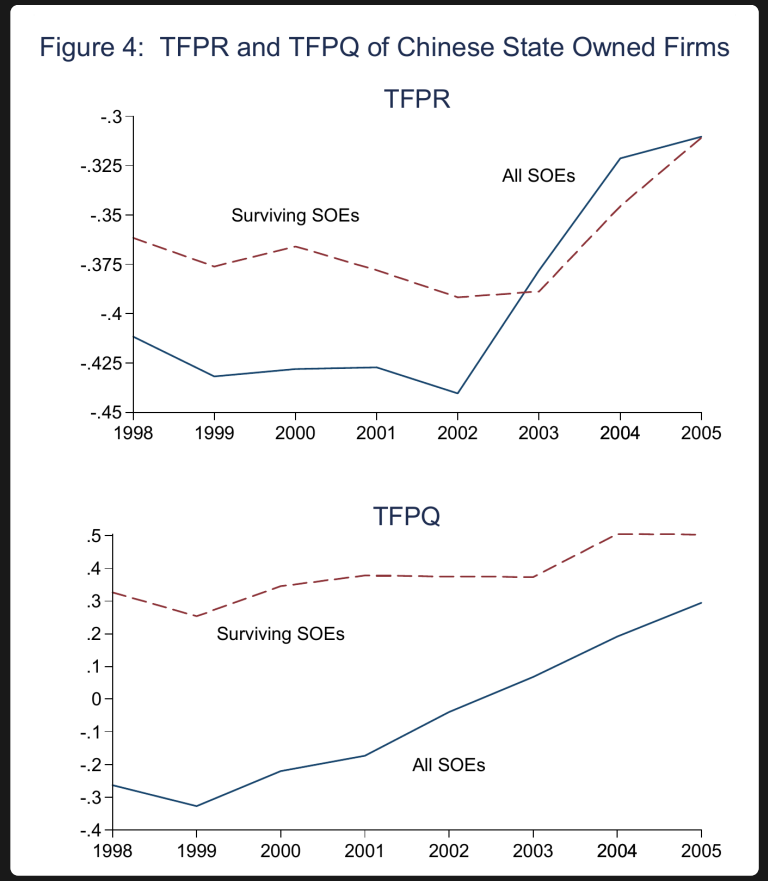


Hsieh and Klenow, Fig. 2 (working paper): TFPR relative to the industry mean, India, China, US

- TFPR, 90th over 10th percentile: **4.9** in China, **5.0** in India, **3.3** in the US
- Make TFPR equal everywhere: Chinese manufacturing TFP **+115%** (1998), **+96%** (2001), **+87%** (2005)
- The US would gain too: **+43%** (1997). Only up to **US efficiency**, China's gain divided by the US gain (2.15 ÷ 1.43): **+50%, +37%, +30%**
- Most firms would **shrink**; a few large, productive ones would grow a lot

RESULTS: THE STATE FIRMS

- State plants have about **40% lower TFPR**, "as if they received subsidies to continue operating despite low profitability"
- Their share of plants falls: **29%** (1998), **19%** (2001), **8%** (2005)
- And the TFPR gap of the survivors **narrows** after 2002 (figure)
- About **4 in 10** of the improvement in allocation between 1998 and 2005 comes from that narrowing gap



Hsieh and Klenow, Fig. 4 (working paper): log TFPR and TFPQ of state firms relative to other firms, 1998-2005

A LEVEL, NOT A GROWTH RATE

- "+30-50%" is a **level**: how much higher TFP would be, in a given year, with a better allocation
- The model is **static**: the wedges are frictions, reallocation happens once. There is nothing in it that changes the growth **rate**
- Growth appears only when the gap **changes**: allocation in China improved by about **15%** from 1998 to 2005, about **2% a year**
- TFP in Chinese industry grew about 6% a year: so "perhaps **one third**" of it came from better allocation, while the gap was closing

ROBUSTNESS

- **Demand**: with $\sigma = 5$ instead of 3, the gains get **bigger**: China in 2005 from +87% to +184%
- **Outliers**: they cut the 1% tails of the distributions before computing anything
- **Technology**: let each plant have its own capital share instead of a capital wedge; the gaps remain
- **Noise**: TFPR is linked to ownership, as policy would predict and noise would not

WHY IT MATTERS

- Part of what makes a country poor is **where** its capital and workers are: good firms too small, bad firms too big
- In China a visible wedge is **state ownership**: cheap credit, soft budgets, workers kept on the payroll
- Between 1998 and 2005 the gap narrowed, as the state firms were closed, sold and reformed. **How**, and what was it worth? Tomorrow

HOMEWORK, FOR TOMORROW

- Think of the economy of the country where you **grew up**: Italy, or any other
- Name at least **three wedges** in it: anything that makes some firms pay less, or more, for capital or workers, or keeps them from growing or shrinking
- For each: who gains, who loses, what it costs the economy
- Is it being **removed**, or getting **stronger**?
- The SOE Lab can help you think. Write your answers down: I will ask you **tomorrow, in class**

TOMORROW: WHAT CHINA DID WITH ITS STATE FIRMS

- **Hsieh and Song (2015)**: "Grasp the large, let go of the small". Who was closed, who was sold, who was kept, and what it was worth
- **Khandelwal, Schott and Wei (2013)**: in 2005 the world's quotas on textiles end. Licences had gone to state firms. What happens when the wedge disappears overnight?

ADDITIONAL MATERIAL: THE FORMULA BEHIND THE GAINS

HSIEH AND KLENOW (2009), EQUATIONS (2.15)-(2.16)

Industry TFP, given the allocation:

$$TFP_s = \left[\sum_i \left(A_i \frac{\overline{TFPR_s}}{TFPR_i} \right)^{\sigma - 1} \right]^{\frac{1}{\sigma - 1}}$$

If TFPQ and TFPR are jointly log-normal:

$$\log TFP_s = \frac{1}{\sigma - 1} \log \left(\sum_i A_i^{\sigma - 1} \right) - \frac{\sigma}{2} \operatorname{var} \left(\log TFPR_i \right) - \frac{\alpha(1-\alpha)}{2} \operatorname{var} \left(\log \frac{MRPK_i}{MRPL_i} \right)$$

- Three terms: how good the firms are; how badly inputs are spread among firms; firms with the wrong mix of capital and workers
- With $\sigma = 3$, the second term is 1.5 times the variance of log TFPR
- Equation (2.16) of the working paper leaves out the third term, which is zero only if all firms have the same ratio of capital wedge to labour wedge